Trend and Seasonal Components Transcript

Hello everyone, welcome to lesson 2 video 2. In this video, we will extend our forecasting methods to include trend and seasonal components. Last time we learned about simple exponential smoothing, which is great for forecasting data that does not have a trend, right? Now what if the data has an upward or downward looking trend? And in those cases, the easy models that we have like exponential smoothing and all, they fail. Now, there are methods like Holt Winters, which we are going to learn in this video and we will get to understand how Holt Winters captures trend and effectively models or helps us model the time series data and gives us more sophisticated forecasting.

kay. Now, the basic idea for Holt's method is to not only have first level forecast, which is being used at smoothing or which is used for smoothing. This is the same equation if you remember that we had for smoothing, but we also have a second equation which gives an estimate of the slope.

Okay. Now, once we have an equation like this, now if you see here, the forecast is measured as y is equals to m into x plus c. This is the equation of the line and this is the slope, right? Sorry, this is the slope bt and bt is estimated using a smoothing method for trend. So, we estimate first the intercept, which is lt, we use that in the equation here and then we use a second equation to estimate the slope as well, which is bt using the second equation here and once we have them, we use the module or method or this equation to give us a forecast.

So, this equation is used to get just the forecast. Generally, these forecasts are good enough to capture the linear upward or downward linear trend. And they can be used for forecasting more complicated times, not complicated, but at least the ones which are having linear upward trend.

Okay. So, this method, as I have told you, this method has two different components. One for level lt, which is the intercept term and then the second term, which is bt, which is the slope term.

Okay. We actually can have a third term, which is seasonal as well. And once we have it, all of these three terms, we can define models like additive model, which we studied in previous section, wherein our time series is decomposed into three components, which were our linear component like trend and then our seasonal component and the component, which has errors, right? The fluctuations or irregularities.

So, those were the three components, t, s and i, trend, seasonality and irregularities. Now, similar to that, we can have, once we have level, trend and seasonal components, we can define an additive time series. Similarly, if the trends are like the cycles move at a gap, or let's say, for example, the cycles width and the cycle increase, in those cases, we have a multiplicative model.

In that case, we multiply the trend with seasonal and the level components that we derive here at the top. Okay. So, this is how we can actually use simple or extend the idea of smoothening to Holtz-Winters method, wherein we not only derive the exponential, simple exponential term, but also derive the trend term, which would help us to understand what does that trend, how does that trend looks like in the linear, entire linear data, or is there any upward trend or downward trend? And once we learn about those trends, it becomes easier for us to use the equation which we learned previously, an equation like this, which is nothing but an equation of a line, right? y is equals to mx plus c, and It is treated as a intercept term, and but here is the slope term for us.

Okay. So, this is the entire method. And we will do some practical examples while doing live classes, so that you get an understanding of the entire method much more, and you have a practical understanding of how it works.

Now, just to help you understand how good they are to estimate the trend, if you see here, there are cyclic pattern, this is an additive model, if you see the cycles are not varying its gap, right, the gap or the width between the cycle is almost constant, it means that it is an additive model. But it also has a trend, right, a trend component, which is increasing upwards. Now, because it has a trend, we cannot use or we, if we use the methods like exponential smoothening or moving averages, they might not do justice to this type of trends, okay.

So what we do, we use Horst's method and Horst's method, they capture trend along with seasonality and if we also derive the irregularity components, we can add them together, because this looks like an additive model. And once we add it, we will get a much more smoothed out curve, because it is a smoothed out curve. Now, if we have a data only up till here, if you think of, and if we have to use the same learned curve to extend the forecast to the future, the right side on the right hand side, we have dashed red curve where my cursor is, these are the forecast.

Now, if you see here, it actually has learned the seasonal component very well, which is what has been, which you can see here, right. It is also learned the linear term as well, because it is, it is following the curve. That's what you can see, right.

So this entire method, Holt-Winters, they work really, really well in understanding the trend component. Plus, if we also add seasonality to it, they will be effective in creating the seasonal effect and they will be able to give us forecast in any time series model. So this is all about Holt-Winters.

I hope the understanding is clear to everyone. Many real-world time series or many real-world time series data show steady growth and repeat seasonal effects.

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